



ME 300: Engineering Thermodynamics

Gibbs Equation of State

Gibbs equations of state

$$\text{S.C.S.} \rightarrow \underbrace{f(\mathcal{P}_1, \mathcal{P}_2)}_{\substack{\text{independent} \\ \text{intensive}}} = \mathcal{P}_3$$

$$\text{P-v-T: } P = f(T, v)$$

$$\text{Caloric: } u = f(T, v) \\ h = f(T, P)$$

$$\text{Gibbs: } \left. \begin{array}{l} S = f(T, P) \\ S = f(T, v) \end{array} \right] \star$$

$g = f(T, P)$ Gibbs Free energy

$$g = h - Ts$$

Deriving Gibbs equations of state

→ First law: $d\bar{U} = \delta Q - \delta \bar{W}$
reversible

$$\Rightarrow \delta Q_{\text{rev}} \Rightarrow dS = \frac{\delta Q}{T} \Big|_{\text{rev}} \rightarrow \delta Q_{\text{rev}} = T dS$$

$$\Rightarrow \delta \bar{W}_{\text{rev}} \Rightarrow \delta \bar{W} = P dV$$

$$\rightarrow \boxed{\begin{aligned} d\bar{U} &= T dS - P dV \\ du &= T ds - P dv \end{aligned}}$$

$$\begin{aligned} H &= \bar{U} + P\bar{V} \rightarrow \bar{U} = H - P\bar{V} \\ d\bar{U} &= dH - P d\bar{V} - \bar{V} dP \end{aligned}$$

$$\boxed{\begin{aligned} dH &= T dS + \bar{V} dP \\ dh &= T ds + v dP \end{aligned}}$$

For ideal gases

$$\rightarrow P_v = RT$$

$$du = C_v dT$$

$$dh = C_p dT$$

$$Tds = du + Pdv$$

$$= C_v dT + Pdv$$

$$ds = C_v \frac{dT}{T} + P \frac{dv}{T}$$

$$ds = C_v \frac{dT}{T} + R \frac{dv}{v}$$

$$\Delta S = \int_{T_1}^{T_2} C_v \frac{dT}{T} + \int_{v_1}^{v_2} R \frac{dv}{v}$$

assume $C_v = \text{const}$

ideal gas
 $C_v = \text{const}$

$$\Delta S = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{v_2}{v_1}\right)$$

$$Tds = dh - v dP$$

$$= C_p dT - v dP$$

$$ds = C_p \frac{dT}{T} - v \frac{dP}{T}$$

$$ds = C_p \frac{dT}{T} - R \frac{dP}{P}$$

$$\Delta S = \int C_p \frac{dT}{T} - R \int \frac{dP}{P}$$

$C_p = \text{const.}$

$$\Delta S = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$$



The change in entropy in an ideal gas for an isothermal process where pressure increases is which of the following?

A. $\Delta S > 0$

B. $\Delta S < 0$

C. $\Delta S = 0$

$$\Delta S = C_p \ln\left(\frac{T_2}{T_1}\right) - \underbrace{R \ln\left(\frac{P_2}{P_1}\right)}_{>0}$$